SCHOOL OF MANAGEMENT

SESSION 2026-27



Sectoral Functional Analytics

(NBBAB5501)

Assignment

SUBMITTED BY - Piyush Maurya

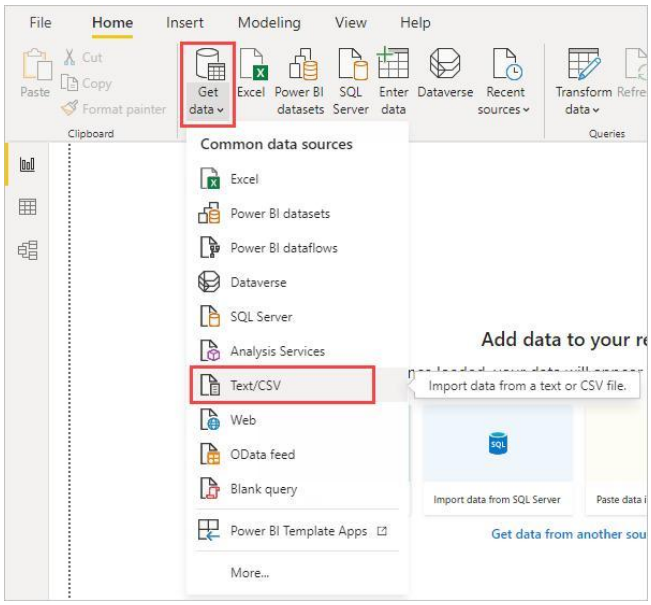
Roll No. - 1240645138

COURSE - BBA(BA)

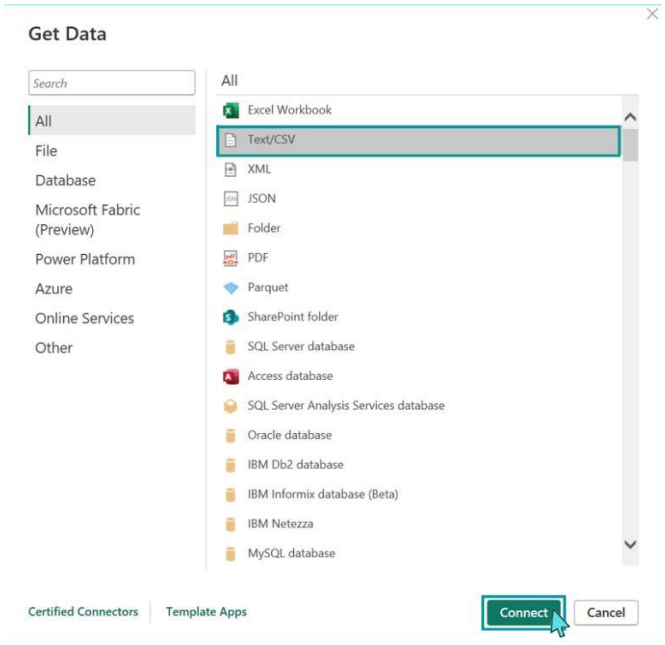
SECTION – BBAB32

SUBMITTED TO – Mr. Arif Raza

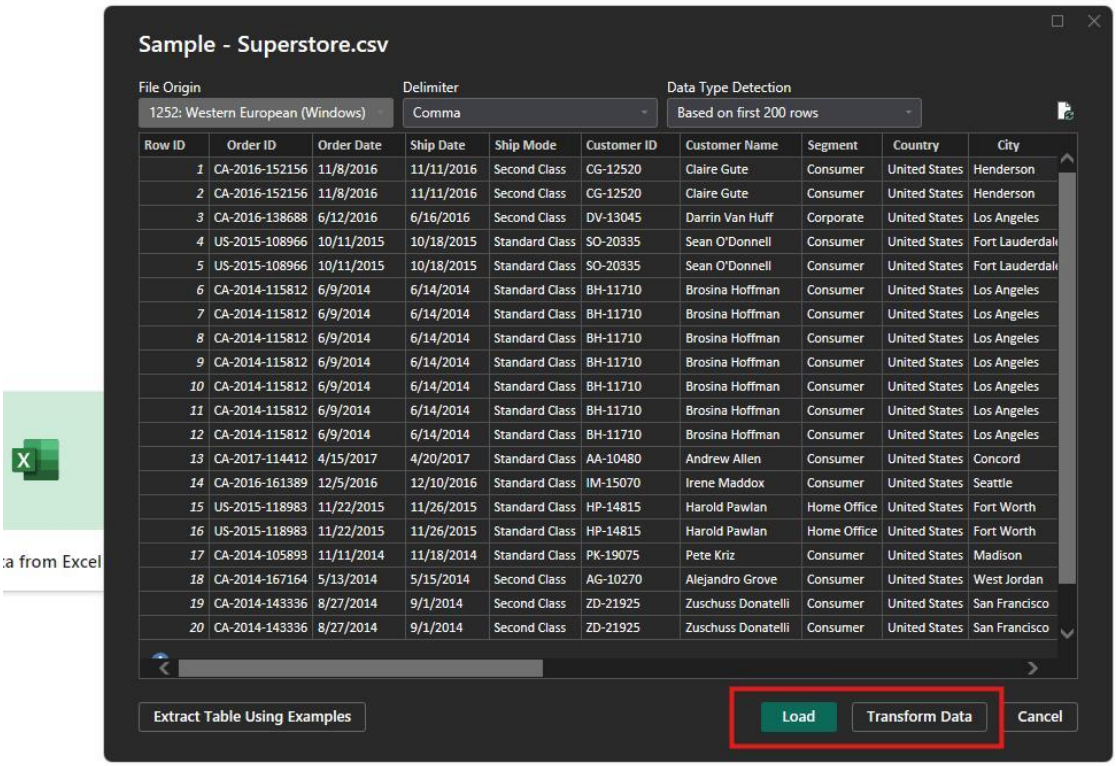
Step 1: Import the CSV



- Open Power BI Desktop
- Open a new blank report.



- Select the CSV
- Go to Home → Get data → Text/CSV. Browse to your downloaded file and select it.



- Open Power Query
- In the preview window, click Transform Data, not Load. This lets you clean the dataset before using it.

Step 2: Check data types

In Power Query Editor, select each column and set the correct data type.

Column	**Data type**
Student_ID	Text
Attendance_Percentage	Whole number
Study_Hours_Per_Day	Decimal number
Previous_Exam_Score	Whole number
Assignments_Completed_Out_Of_10	Whole number
Internal_Marks_Out_Of_100	Decimal number
Pass_Fail	Text

Step 3: Check missing values and duplicates

1. Select View → Column quality to display valid, error, and empty values.
2. Check whether any cells contain null or errors.
3. Select Student_ID.
4. Go to Home → Remove Rows → Remove Duplicates only if duplicate IDs are confirmed.
5. Check that attendance is between 0 and 100 and assignment counts are between 0 and 10.

Do not delete records just because their marks or attendance are low. Those records may be important for predicting failures.

Once finished, click Home → Close & Apply.

Step 4: Create DAX measures

In Power BI, go to Modeling → New measure and create these measures one at a time.

```
-Total Students =
DISTINCTCOUNT(
    student_pass_fail_prediction[Student_ID]
)
------------------------------------------------------------
-Passed Students =
CALCULATE(
    [Total Students],
    student_pass_fail_prediction[Pass_Fail] = "Pass"
)
------------------------------------------------------------
-Failed Students =
CALCULATE(
    [Total Students],
    student_pass_fail_prediction[Pass_Fail] = "Fail"
)
```

-Pass Percentage =

```
DIVIDE(
    [Passed Students],
    [Total Students],
    0
)
```

Step 5: Add the visuals

- Click an empty area on the report canvas.
- Select the Card visual and drag Total Students into its data field.
- Add a second Card using Pass Percentage.
- Select a Donut chart. Drag Pass_Fail into Legend and Student_ID into Values, using Distinct count.
- Select a Clustered column chart. Put Pass_Fail on the X-axis and Attendance_Percentage on the Y-axis; change the aggregation to Average.
- For study hours and internal marks, repeat with their respective columns and average aggregation.
- Add a Slicer visual and drag Pass_Fail into it

Step 6: Complete the prediction model

Your SPSS Modeler workflow will be:

CSV → Type → Partition → C5.0 → Analysis/Evaluation

- Set Pass_Fail as the target.
- Set the student characteristics as input fields.
- Use a training/testing split.
- Evaluate the model with a confusion matrix, accuracy, precision, recall and F1-score.
- If your goal is to predict failure before final results, exclude any features that would only be known after the outcome, such as final exam marks.

STUDENT PERFORMANCE ANALYSIS
Pass/Fail Prediction Project

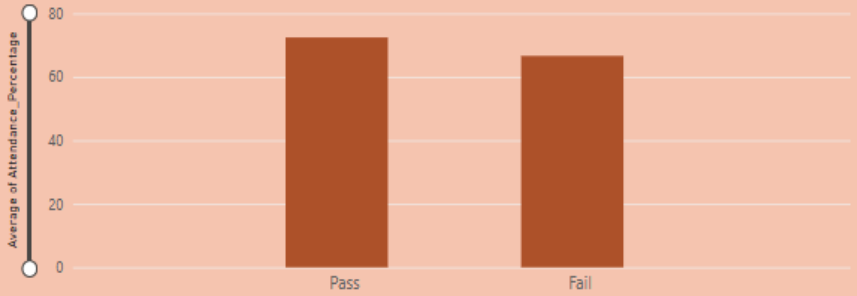
Total Student

Total Students
200

Passed Students

Passed Students
180

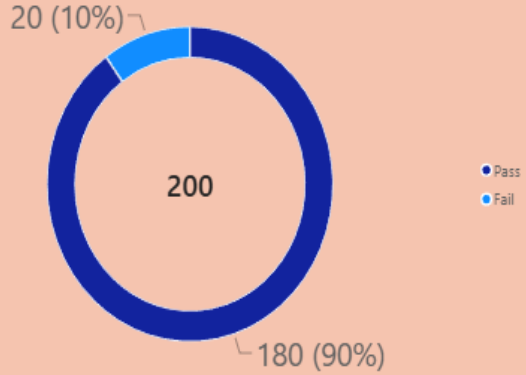
Average of Attendance_Percentage
by Pass_Fail



Pass_Fail

Enter Pass_Fail

Count of Student_ID
by Pass_Fail



Average of Study_Hours_Per_Day and Average of Internal_Marks_Out_Of_100
by Student_ID

